CHEMISTRY ONLY

# 26 ALCOHOLS

Alcohols are an important class of organic compounds. In this chapter we will study some of their reactions.



▶ Figure 26.1 Alcohols are very useful solvents.

## LEARNING OBJECTIVES

- Know that alcohols contain the functional group –OH
- Understand how to draw structural and displayed formulae for methanol, ethanol, propanol (propan-1-ol only) and butanol (butan-1-ol only), and name each compound
- Know that ethanol can be oxidised by:
  - burning in air or oxygen (complete combustion)
  - reaction with oxygen in the air to form ethanoic acid (microbial oxidation)
  - heating with potassium dichromate(VI) in dilute sulfuric acid to form ethanoic acid
- Know that ethanol can be manufactured by:
  - reacting ethene with steam in the presence of a phosphoric acid catalyst at a temperature of about 300 °C and a pressure of about 60–70 atm
  - the fermentation of glucose, in the absence of air, at an optimum temperature of about 30 °C and using the enzymes in yeast
- Understand the reasons for fermentation in the absence of air and at an optimum temperature

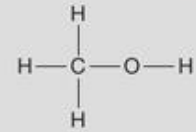
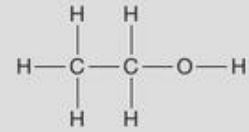
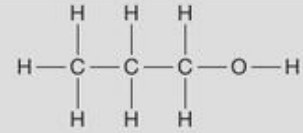
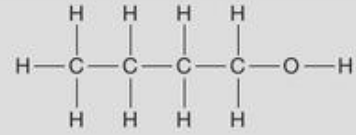
What we commonly refer to as 'alcohol' is just one member of a large family (homologous series) of similar compounds. Alcohols all contain the functional group **–OH** covalently bonded to a carbon chain.

The familiar alcohol in drinks is $C_2H_5OH$ (or, better, showing the structure, $CH_3CH_2OH$) and should properly be called *ethanol*.

## DRAWING AND NAMING THE ALCOHOLS

We are only going to look at the four simplest alcohols: methanol, ethanol, propan-1-ol and butan-1-ol. These are shown in Table 26.1.

Table 26.1 Formulae and names of the first four members of the alcohol homologous series

| Name | Molecular formula | Structural formula | Displayed formula |
|---|---|---|---|
| methanol | $CH_4O$ | $CH_3OH$ |  |
| ethanol | $C_2H_6O$ | $CH_3CH_2OH$ |  |
| propan-1-ol | $C_3H_8O$ | $CH_3CH_2CH_2OH$ |  |
| butan-1-ol | $C_4H_{10}O$ | $CH_3CH_2CH_2CH_2OH$ |  |

**HINT**

When drawing the displayed formula for alcohols you must show the bond between the O and the H, so O–H.

Look carefully at the names. The 'meth', 'eth', 'prop' and 'but' parts count *the number of carbon atoms*. The 'an' part tells you that the molecules are *saturated* and there aren't any carbon–carbon double bonds. The 'ol' tells you there is *an –OH functional group*.

Why the number 1 in propan-1-ol and butan-1-ol? The number tells you *where the –OH group is in the carbon chain*. You count the numbers from the end which gives you the smallest possible number in the name (see Table 26.1). Here, because the –OH group is attached to the end carbon, it is on carbon number 1. It is also possible to have propan-2-ol or butan-2-ol if the –OH is attached to a middle carbon. Propan-1-ol and propan-2-ol are structural isomers of each other, the same holds for butan-1-ol and butan-2-ol.

**HINT**

The names propanol and butanol are acceptable to use in the exam.

For the rest of this topic, we are just going to use ethanol as a representation of alcohols.

## THE OXIDATION OF ETHANOL

### ETHANOL BURNS IN AIR

All alcohols burn to form carbon dioxide and water. With ethanol:

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$$

Ethanol is a **biofuel**, that is, *a fuel that is made from biological sources, such as sugar cane or corn*. Mixtures of petrol with ethanol are increasingly used in countries such as Brazil, which have little or no oil industry to produce their own petrol. On the other hand, they often have a climate which is good for growing sugar cane. Other countries are introducing biofuels such as ethanol to reduce dependence on fossil fuels, which are finite **non-renewable resources**.



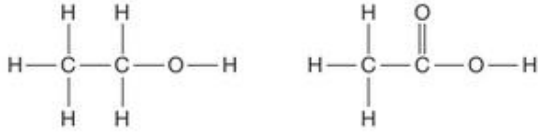

▲ Figure 26.2 Ethanol being used as a fuel in Brazil.

## ETHANOL IS OXIDISED BY THE AIR IN THE PRESENCE OF MICROBES (MICROBIAL OXIDATION)

A bottle of wine left open to the air turns sour. The French for sour wine is *vin aigre*, which has been distorted into *vinegar*. The ethanol in the wine is oxidised by air with the help of microorganisms such as bacteria or yeast to form ethanoic acid, $CH_3COOH$.

**DID YOU KNOW?**

The old name for ethanoic acid was acetic acid.




▲ Figure 26.3 Displayed formulae of ethanol and ethanoic acid

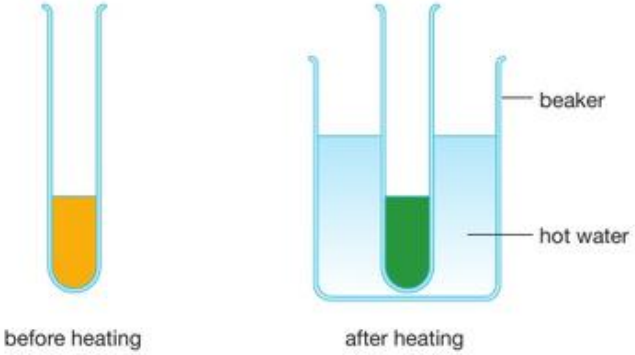
## ETHANOL CAN BE OXIDISED BY HEATING WITH POTASSIUM DICHROMATE(VI) IN DILUTE SULFURIC ACID

**EXTENSION WORK**

In the name potassium dichromate(VI), the 'di-' tells you that there are two chromium atoms in the compound, and the roman numeral (VI) shows the *oxidation state* of chromium in this compound. We have seen these symbols before in Chapter 7, where they simply represent the charges on the ions, i.e. Cu(II) is $Cu^{2+}$ and Fe(III) is $Fe^{3+}$. The situation in the dichromate ion, $Cr_2O_7^{2-}$, is a little more complicated as there is covalent bonding between the chromium and oxygen atoms within the ion. You do not need to worry about the concept of oxidation state at International GCSE. However, it is important to add the (VI) after the name of potassium dichromate as there are other types of chromates which cannot oxidise alcohols.

This is the usual way of oxidising ethanol (and other alcohols) in the lab. The oxidising agent is a mixture of potassium dichromate(VI) and dilute sulfuric acid. Potassium dichromate(VI) is a strong oxidising agent and has the formula $K_2Cr_2O_7$. The dilute sulfuric acid is important for the potassium dichromate(VI) to act as an oxidising agent. Without the $H^+$ ions from the acid, no redox reaction will occur.

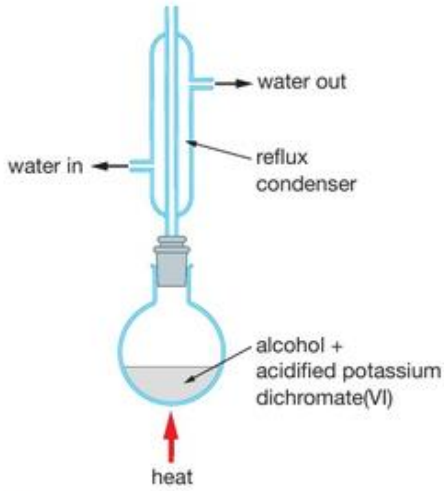
A few drops of ethanol are added to a solution containing the orange mixture of potassium dichromate(VI) and dilute sulfuric acid in a test-tube. The tube is heated in a hot water bath.



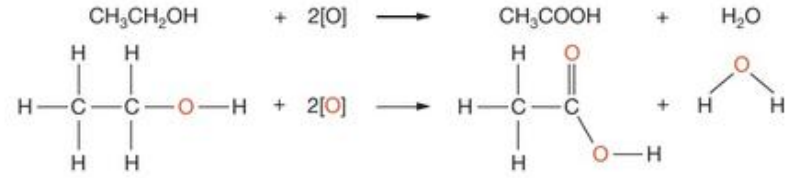

**!** Safety Note: Wear eye protection and add the reactants very slowly to prevent too violent a reaction. Do not point the mouth of the test-tube towards you as the contents may 'spit' out.

▲ Figure 26.4 Oxidation of ethanol by heating with potassium dichromate(VI) and dilute sulfuric acid.

The solution turns green and now contains a very dilute solution of ethanoic acid together with other products. The green colour indicates the presence of $Cr^{3+}$ ions, which are formed when the potassium dichromate(VI) is reduced. The equation for this reaction is very complicated, and so in organic chemistry we frequently show an oxidising agent in equations as [O].




▲ Figure 26.6 Ethanol is refluxed with acidified potassium dichromate(VI) to form ethanoic acid.




Starting materials: ethanol and potassium dichromate(VI) in dilute sulfuric acid

Conditions: heat under reflux

▲ Figure 26.5 Oxidation of ethanol to ethanoic acid

**EXTENSION WORK**

You can use this reaction to make a sample of ethanoic acid by heating ethanol with an excess of the oxidising agent in a flask with a condenser placed vertically in the top (Figure 26.6). This is known as *heating under reflux* and condenses any vapours so that they run back into the flask and don't escape.

When the reaction is complete, the condenser is rearranged for a simple distillation. Ethanoic acid and water are both distilled and a solution of ethanoic acid is collected.

## THE PRODUCTION OF ETHANOL

Here we will look at two different process that are used industrially to make ethanol.

### MAKING ETHANOL BY FERMENTATION



▲ Figure 26.7 Sugar cane: one of the possible raw materials for making ethanol. Others include maize (called corn in the USA), wheat and other starchy materials.

Yeast is added to a sugar (or starch) solution and left in the warm (about 30 °C) for several days in the absence of air (anaerobic conditions). **Enzymes** (biological catalysts) in the yeast convert the sugar into ethanol and carbon dioxide. This process is known as **fermentation**.

The *absence of air* and the *temperature* are both important. In the presence of air (aerobic conditions), enzymes in the yeast produce carbon dioxide and water instead of ethanol. The enzymes are protein catalysts and if the temperature is increased much above 40 °C they lose their structure and don't work any longer. The proteins are said to be **denatured**. At a lower temperature, the reaction is too slow. Somewhere between 30 and 40 °C is the *optimum temperature* for the reaction.

**EXTENSION WORK**

The biochemistry is very complicated. Assuming you are starting from ordinary sugar (sucrose), the sucrose is split into two smaller sugars, glucose and fructose. Glucose and fructose have the same molecular formulae, but different structures. They are structural isomers.

$$\underset{\text{sucrose}}{C_{12}H_{22}O_{11}(aq)} + H_2O(l) \rightarrow \underset{\text{glucose}}{C_6H_{12}O_6(aq)} + \underset{\text{fructose}}{C_6H_{12}O_6(aq)}$$

Enzymes in the yeast convert glucose into ethanol and carbon dioxide in a large number of steps. The overall equation for that conversion is

$$\underset{\text{glucose}}{C_6H_{12}O_6(aq)} \rightarrow \underset{\text{ethanol}}{2C_2H_5OH(aq)} + \underset{\text{carbon dioxide}}{2CO_2(g)}$$

Starting materials: glucose (from sugar cane)
Temperature: 30 °C
Catalyst: enzymes in yeast
Other conditions: anaerobic

**REMINDER**

You will find a diagram and discussion for a lab-based fractional distillation of an ethanol/water mixture in Chapter 2, page 18.

Yeast is killed by more than about 15% of alcohol in the mixture, and so it is impossible to make pure alcohol by fermentation. The alcohol is purified by fractional distillation. This takes advantage of the difference in boiling point between ethanol and water. Water boils at 100 °C whereas ethanol boils at 78 °C.

## MAKING ETHANOL BY THE HYDRATION OF ETHENE

Ethanol is also made by reacting ethene with steam, a process known as hydration.

$$CH_2{=}CH_2(g) + H_2O(g) \rightarrow CH_3CH_2OH(g)$$

| | |
|---|---|
| Starting materials: | ethene and steam |
| Temperature: | 300 °C |
| Pressure: | 60–70 atmospheres |
| Catalyst: | phosphoric acid ($H_3PO_4$) |

**KEY POINT**

This is an *addition reaction*: the water is added to the ethene without removing anything.

Only a small proportion of the ethene reacts. The ethanol produced is condensed as a liquid and the unreacted ethene is recycled through the process.

## COMPARING THE TWO METHODS OF PRODUCING ETHANOL

| | Fermentation | Hydration of ethene |
|---|---|---|
| Use of resources | Uses renewable resources: sugar beet or sugar cane, corn and other starchy materials | Uses finite, non-renewable resources: once all the oil has been consumed, there won't be any more |
| Type of process | A batch process: everything is mixed together in a reaction vessel and then left for several days<br>That batch is then removed and a new reaction is set up<br>This is inefficient | A continuous flow process: a stream of reactants is constantly passed over the catalyst<br>This is more efficient than a batch process |
| Rate of reaction | Slow, taking several days for each batch | Quick |
| Quality of product | Produces very impure ethanol which needs further processing | Produces much purer ethanol |
| Reaction conditions | Uses gentle temperatures and atmospheric pressure | Uses high temperatures and pressures, requiring a high input of energy |

At the moment, countries which have easy access to crude oil produce ethanol mainly from ethene, but one day there will be no oil remaining. At that point, the production of ethanol from sugar and starch will provide an alternative route to many of the organic chemicals we need.